

Milestone4 Progress Report

MedAssist AI: Medical Symptom Analysis & Disease Prediction System

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Project Title: MedAssistAI – Medical Symptom Analysis & Disease Prediction System

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1.Introduction

This report explains the work completed in Milestone 4 of the MedAssist AI project.

The main goal of this milestone was to move the application from the local computer to a cloud server. The React frontend and FastAPI backend were deployed using Render.

This milestone also included completing the user interface, connecting the frontend with the AI model, testing the prediction system, generating medical recommendations, and creating PDF reports.

2. Cloud Deployment Architecture

The MedAssist AI application was deployed using Render. The frontend and backend were deployed separately.

The React frontend is used by the patient, while the FastAPI backend handles the API requests, data processing, and AI prediction.

2.1 Live Deployment Endpoints

The application is available at the following links:

Frontend:

<https://medassist-ai-frontend.onrender.com>

Backend API:

<https://medassistai-unpq.onrender.com>

Swagger API Documentation:

<https://medassistai-unpq.onrender.com/docs>

The frontend is used to interact with the application. The backend receives the user data and sends it to the XGBoost model for prediction.

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2.2 Infrastructure Details

The backend contains the FastAPI application, data-processing code, and trained XGBoost model.

The frontend is deployed as a React application.

Environment variables are used to store important configuration details such as database information and security keys. This helps keep sensitive information safe and prevents it from being directly written in the source code.

The basic working flow is:

User → Frontend → Backend → XGBoost Model → Prediction → Result

3 User Interface Implementation

The user interface was completed during this milestone.

The main aim was to make the application simple and easy for patients to use. The frontend communicates with the backend and displays the results returned by the AI system.

3.1 Patient Dashboard

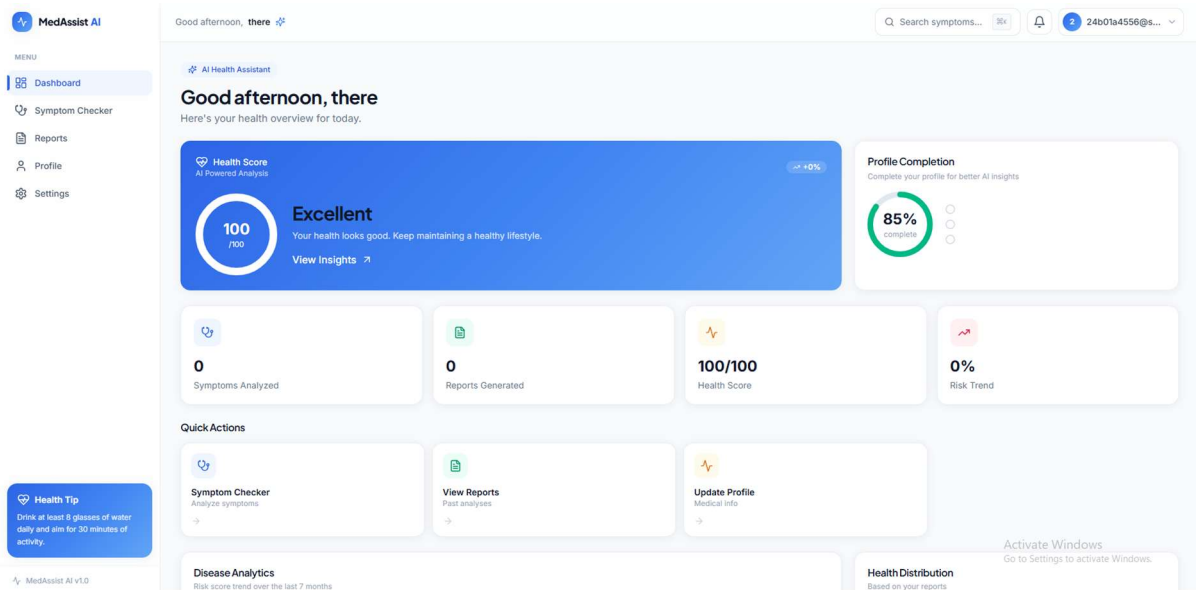
After login, the patient is taken to the main dashboard.

The dashboard contains the following features:

- **Health Score:** Shows the user's health score and basic AI insights.
- **Profile Completion:** Shows how much of the patient's profile has been completed.
- **Metric Cards:** Shows information such as symptoms analyzed and reports generated.
- **Risk Trends:** Shows information about the user's risk levels.
- **Health Tips:** Displays simple health and wellness tips.

For the test user, the profile completion was shown as **85%**.

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3.2 Interactive Symptom Checker

The Symptom Checker is one of the main features of the application.

It allows the user to select symptoms and provide additional information about them.

Step 1 – Symptom Selection

The user can search for symptoms or select them from different categories.

Examples of categories include:

- Cardiology
- Pulmonology
- Dermatology
- Other medical categories

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MedAssist AI

Good afternoon, there

Search symptoms...

MENU

- Dashboard
- Symptom Checker
- Reports
- Profile
- Settings

Symptom Checker

Let's analyze your symptoms

Tell us how you're feeling and our AI will assess possible conditions.

Step 1 of 3 33%

Search symptoms (e.g. headache, fever)...

General Cardiology Pulmonology Dermatology Gastroenterology ENT

Pain & Discomfort Mental Health Gynecology Musculoskeletal Ophthalmology

Urology

☐ abdominal distention	☐ abnormal involuntary movements	☐ absence of menstruation
☐ abusing alcohol	☐ allergic reaction	☐ ankle stiffness or tightness
☐ ankle swelling	☐ ankle weakness	☐ apnea
☐ arm cramps or spasms	☐ arm lump or mass	☐ arm stiffness or tightness
☐ arm swelling	☐ arm weakness	☐ bedwetting
☐ bleeding gums	☐ bleeding in mouth	☐ bleeding or discharge from nipple
☐ blindness	☐ bumps on penis	☐ chills

Health Tip

Drink at least 8 glasses of water daily and aim for 30 minutes of activity.

MedAssist AI v1.0

Step 2 – Duration and Severity

In the second step, the user provides more information about the symptoms.

The user selects:

- Duration of the symptoms
- Severity of the symptoms

Duration options include:

- Less than 1 day
- 1–3 days
- More than 1 week

Severity options include:

- Mild
- Moderate
- Severe

After the user submits the information, it is converted into JSON format and sent to the backend.

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Symptom Checker

Let's analyze your symptoms
Tell us how you're feeling and our AI will assess possible conditions.

Step 2 of 3 67%

How long have you had these symptoms?

< 1 day 1-3 days 4-7 days > 1 week

How severe are your symptoms?

Mild
Noticeable but not disruptive

Moderate
Affects daily activities

Severe
Significantly impacting life

← Back Continue →

4. AI Prediction and Reporting

The AI prediction system was tested using the live application.

When the user submits symptoms, the frontend sends the information to the FastAPI backend. The backend processes the information and sends it to the trained XGBoost model.

The model then provides a prediction. The backend also generates the risk level, confidence score, recommendations, and suggested tests.

The process is:

Symptoms → Backend → XGBoost Model → Prediction → Recommendations → PDF Report

4.1 Diagnostic Case Study: Acute Anxiety Episode

A test was performed on **August 20, 2026** to check whether the complete system was working correctly.

The following six symptoms were entered:

1. Chest tightness
2. Palpitations
3. Breathing fast
4. Difficulty breathing
5. Diarrhea
6. Incontinence of stool

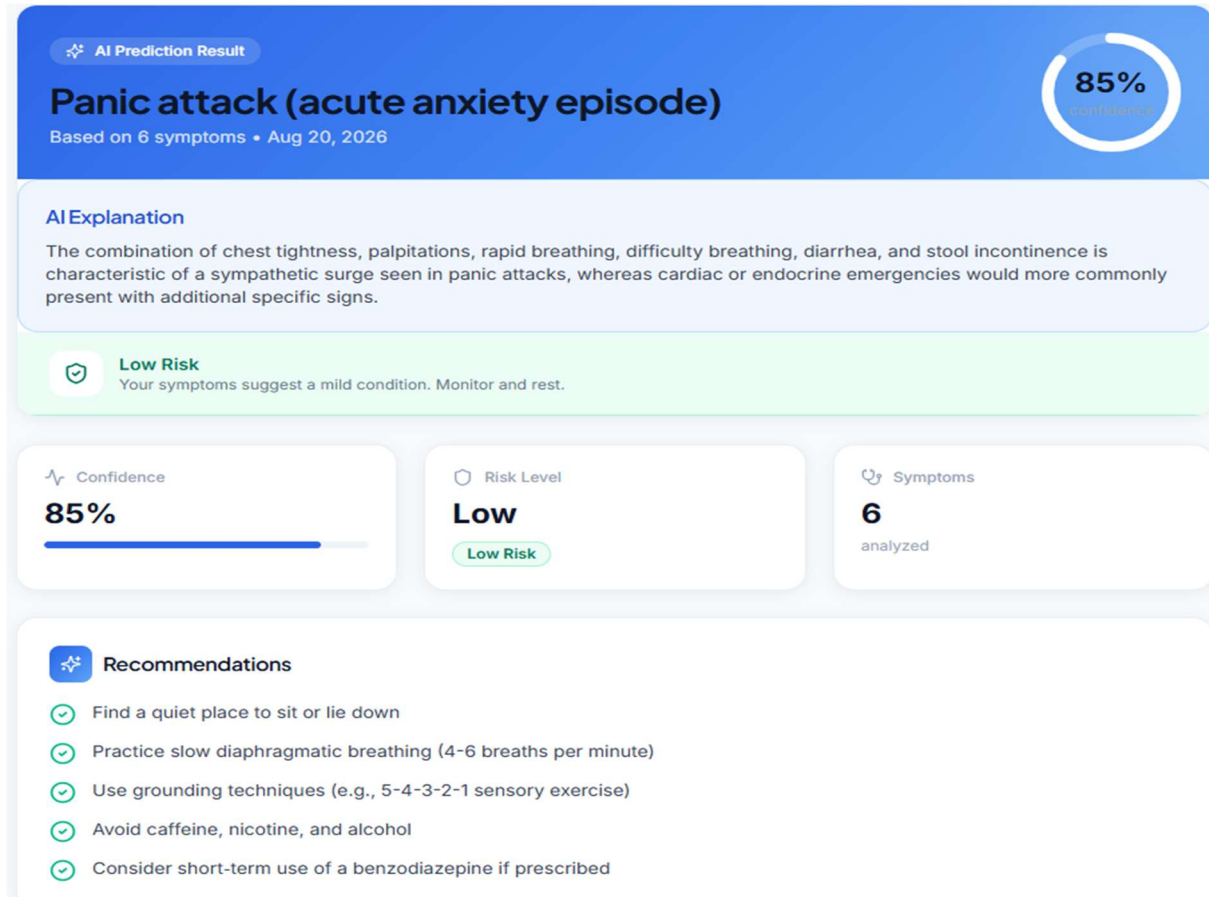
AI Result:

The AI system produced the following result:

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- **Predicted Disease:** Panic attack (acute anxiety episode)
- **Confidence:** 85%
- **Risk Level:** Low

The result was successfully displayed on the frontend.



4.2 Medical Recommendations and PDF Generation

After making the prediction, the system generated recommendations based on the result.

For this test case, the recommendations included:

- Practice slow breathing.
- Sit or lie down in a safe place.
- Use grounding techniques.
- Avoid caffeine, nicotine, and alcohol.
- Consult a medical professional when necessary.

The system also suggested some follow-up tests:

- **ECG (Electrocardiogram)**
- **Basic Metabolic Panel**

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The system successfully generated a PDF report containing the prediction, confidence score, risk level, recommendations, and medical disclaimers.

Report ID:

3dba826a-4c47-4586-a7aa-37fd27aa06cb

5. Conclusion

Milestone 4 was successfully completed.

The MedAssist AI application was deployed to the cloud using Render. The React frontend and FastAPI backend were successfully connected with the XGBoost machine learning model.

The patient dashboard and Symptom Checker were also completed. The system was tested using a real example of multiple symptoms and successfully produced an AI prediction with a confidence score and risk level.

The application can also generate medical recommendations and downloadable PDF reports.

In the future, the project can be improved by training the model with more data, improving prediction accuracy, adding more diseases and symptoms, and expanding the recommendation system.